Disposition of the Consolidated Manuscript Error and Revision Register

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What this document is

The register of 11 October 2026 lists 52 items. This document records the disposition of each, and sets out the reasoning for every item that was *not* applied as written. It covers Draft 17 of *Chemistry-resolved disagreement between PBE and r²SCAN stability verdicts in an AI-generated materials database*.

Of the 52 items, 45 are applied in full, 6 are applied in part, and 1 cannot be done by the author alone. Nothing is declined on the ground that the register is wrong about the facts. Where an item is not applied as written, it is for one of three reasons: the register itself offered an alternative and the alternative was taken; the work the item asks for would churn every table in the paper to change a number in the fourth significant figure; or the item asks for something only a second person can supply.

Six items were resolved by measurement rather than by rewording, and in two cases the measurement contradicted what the paper previously said. Those are listed in Section 3 because they are the places where the register changed a result and not only a sentence.

1 Items not applied as written

Item 01 — the register's second option was taken, not its first

The register offers two routes. The first regenerates every full-release analytical output at $n = 139{,}516$; the second withdraws the universal-exclusion claim and labels the retained outcomes correctly.

Route two was taken, and the reason is arithmetic. The 25 matched entries that were not PBE-stable were counted: 24 of them fall inside a primary composition cell. Excluding them moves the release-based rate from 13.0413% to 13.0401% and the primary-cell rate from 12.9150% to 12.9136%. Regenerating 40 cell rates, 83 element rates, every cluster bootstrap interval, the regression, the band-gap table and the budget evaluation to move the third decimal place of two percentages would create a large diff with no substantive change, and would break the correspondence between the paper's n and the released file that readers will download.

Instead Section 3.4 now defines two outcomes separately — *release-based disagreement*, which makes no use of the PBE verdict and is defined for all 139,541 entries, and a *confirmed transition*, which requires an observed non-positive PBE decomposition energy and is defined for the 117,018 matched PBE-stable entries. The 25 are excluded wherever a confirmed transition is the outcome or an observed PBE energy is required (the budget evaluation, the propensity-weighted rates, the energy-displacement analysis, the matched comparison) and retained in the release-based rates and the cell tabulations. Both rates are printed, with and without them. The universal-exclusion claim is gone. The population-flow

table now marks which rows are raw release counts and which are analytical, and `run_all.py` checks the 25 and the 24 as headline quantities so the distinction cannot silently rot.

Item 10 — seed stability was demonstrated instead of raising the replicate count

The register permits either. Raising 300–600 replicates to, say, 5,000 would multiply the runtime of the whole bootstrap stack, and would not answer the question the item actually raises, which is whether the *thresholded count* of significant contrasts is stable. So the count was recomputed under five seeds: 53, 53, 52, 53, 53. That is reported in Appendix C and checked in CI as two headline quantities (minimum 52, maximum 53). If the range had been wider the replicate count would have had to go up; it is not.

Item 11 — the reduced-stoichiometry sensitivity check was not run

The register marks it optional. The limitation is now stated explicitly in the paper: compositions are used exactly as released and are not reduced, so the clustering key depends on the released unit-cell convention as well as on the underlying stoichiometry, and the same compound released as A_2B_4 and as AB_2 would receive different anonymised patterns. The space-group source is named and the absence of missing values recorded.

Running the check would mean re-deriving all 8,429 cluster keys and re-running every cluster bootstrap, cluster-robust regression and cluster-wise budget split on the new keys. That is a day of compute and a second set of every interval in the paper, to test a proxy that the paper already declines to claim a direction for. It is recorded as the obvious next robustness check rather than done.

Item 47 — independent review of the core definitions and scripts is not something I can supply

Everything mechanical in this item is done: the pipeline was regenerated on final inputs, script-to-output mappings were checked, the CI checks were extended from 23 headline quantities to 38 (including the updated Benjamini–Hochberg count, the analytical population counts and the new budget contrasts), the environment is pinned, and SHA-256 checksums of both input files are now printed in Appendix F.

The last clause — *obtain independent review of core definitions and scripts* — cannot be satisfied from inside the project. The Data and code availability section now says so in as many words: the core definitions and scripts have not been reviewed by anyone other than the author. This is the one item on the register that is open, and it is the one the accompanying strategic note is right to call the highest-value remaining step.

Item 51 — the citation ledger is partial, and the disclosure now says exactly how partial

Six sources were checked against their source text rather than against the citation, and the checks are recorded in the Disclosures section: Kingsbury et al. (6,307 materials behind "approximately 6000 solids"; r^{2}SCAN more accurate than both SCAN and PBEsol for formation energies); Kothakonda et al. (GGAs comparable for transition metals, intermetallics and weakly bound solids; PBE slightly more accurate for the 233 Type 2 decomposition enthalpies); An et al. (603 to 298, 166 dynamically stable, 109 surviving, SCAN and not r^{2}SCAN, and the exclusion of 142 rare-earth compounds for f-electron and pseudopotential reasons); Yu et al. (19,998 unary and binary materials, per-element formation-energy error); Huang et al. (380,190 paired MatPES entries, and thermodynamic validation on exactly

six binary systems: Fe–O, Mn–O, Ti–Cl, Mo–Cl, V–S, Cu–S); and Liu and Meng, whose "more than 60% metallic" figure is now cited as reported and used in no analysis.

The remaining references were verified for existence, authorship and venue but not claim by claim, and updated publication metadata was not re-checked for all of them. The paper does not claim otherwise. Completing the ledger is a reading task on roughly 25 papers; it is listed in Section 2.

Item 52 — the abstract was shortened; the supplement migration and the contents page were not done

The abstract now fits on one page. Repeated defensive phrasing was cut in several places, and the rhetorical formulations the register names ("not additive bookkeeping", "does not generate", "strongest predictor in every specification") are gone from the abstract, results and conclusion.

Two sub-requests were not acted on. The design-effect arithmetic stays in the main text, because the whole inferential apparatus of the paper rests on the claim that these observations are not independent, and a reader who cannot follow that arithmetic in place has no reason to accept any interval in the paper. The table of contents stays, because the paper now carries eight appendices and a reader looking for the input manifest or the all-cells table needs a way to find them.

The paper is also longer than the 39 pages the register saw, not shorter: it is 46. The additions are the weight diagnostics the register demands in item 32, the all-cells supplement it demands in item 38, the input manifest it demands in item 06, the model specification it demands in item 16, and the stronger budget comparison. Those cost about six pages between them. Length was traded for the evidence the register asked for; the trade seems like the right one, but it is a trade and not a clean win.

2 Items the register asked for that only you can do

None of these is a judgement call. They need an account, a person, or an email that is not mine to send.

1. **Independent computational review** (item 47, and §2 of the strategic note). Give a reviewer the repository, the checksums, the pinned environment and the README, and ask for two separate things: whether the central tables regenerate on a clean machine, and whether the eligibility definitions, group encodings, bootstrap contrasts and budget splits are *correct*. The second is not the same as the first and is the one that matters.

2. **Ask the GNoME authors what the "All" column is** (item 41, §3 of the strategic note). The paper's reading of it is marked as an inference throughout and every conclusion depending on it is flagged conditional, which is the best that can be done without an answer. One email settles it.

3. **Deposit the repository in a DOI-minting archive** (Zenodo or similar) and cite the DOI. The paper currently says the deposit has not been made.

4. **Complete the citation ledger** (item 51) on the roughly 25 references not yet checked claim by claim.

5. **Regenerate the Materials Project API key.** It was partly visible in a screenshot during this work.

6. **Make the repository public** before submission. The paper gives its URL.

3 Where the register changed a result, not a sentence

Six items were answered by running something. Two of them contradicted the paper.

Item 18 falsified a caption. The caption of the logistic-regression table claimed that dropping hydrides "moves no coefficient in the third decimal". Measured, it moves the U-list odds ratio from 3.448 to 3.468 — the third decimal of the largest term in the table. The caption now prints all three measured shifts. The claim was wrong and had never been computed.

Item 32 found that a trimming choice was load-bearing and undocumented. Appendix C said the inverse-probability weights were "untrimmed". The code floors the fitted propensity at 10^{-2}. That is not a cosmetic difference: 26 eligible recomputed entries fall below the floor, and leaving them untrimmed drives the largest weight to 6.6×10^5, collapses Kish's effective sample size from 83,875 to 5, and moves the weighted rate from 12.71% to 3.05%. The reported 12.7% is conditional on trimming, and the paper now says so, reports the full diagnostics (propensity quantiles, overlap between recomputed and unrecomputed, maximum weight, effective sample size, share of weight in the heaviest 1%) and shows the estimate is stable at 12.71%, 12.74% and 12.76% for floors of 10^{-2}, 2×10^{-2} and 5×10^{-2}.

Item 37 replaced a hand-wave with a standardisation. The aggregate inversion was previously explained by a commentary's "more than 60% metallic" figure. The anion-class mix is now measured directly: the release is 38.9% intermetallic and 9.8% oxide. Over the twenty matched cells, standardising both sides to the known-materials cell mix, GNoME disagrees *more* by +1.2 points [+0.5, +1.9]; at the GNoME mix, by +1.8. The direction of the aggregate comparison therefore reverses once the mix is held common, which identifies composition as a sufficient explanation of the inversion over those cells.

Item 40 was answered by deletion, with evidence. The claim that known-materials cells concentrate within 20 meV/atom of the hull is gone. The Materials Project `energy_above_hull` field is floored at zero and reads zero for 21.8% of the 152,426 entries pulled, so those margins are not recoverable from it. The paper now says that, and keeps the GNoME-side near-hull gradient as an internal descriptive result only.

Item 43 corrected a cluster count. The main text said half of 8,429 clusters. The code applies eligibility first; the eligible pool is 7,822 clusters and 3,911 are drawn per split. Both numbers are now in the main text and both are checked in CI.

Item 44 was tested rather than hedged. The register asked for the upper-estimate language to be removed because the direction of generated-regressor bias is not established. It is now removed, and the question was also settled empirically. On 20 fresh cluster-wise splits, a hull-distance model with twenty quantile-bin dummies gains nothing over simple closest-first ranking (-0.06 points at a 20% budget, ahead in 3 splits of 20); the combined policy beats that flexible model by +6.00 points (20 of 20); and cross-fitting the cell rates leaves the margin at +6.31 points (20 of 20). Composition still adds after hull distance is modelled nonparametrically, and the gain is not generated-regressor optimism.

4 Verification state of Draft 17

Two layers, both green:

- `run_all.py` regenerates the results directory from the two input CSVs and checks 38 headline quantities, up from 23. Four are claim-level rather than value-level: the budget ordering holds at every budget; the combined policy's margin over the flexible hull model stays positive under cross-fitting; the iron excess is smallest in oxides of all anion classes; composition adds after hull distance is modelled nonparametrically.

- `scripts/check_paper.py` parses every number out of the LaTeX tables and diffs 232 printed values against the result files.

Six new analysis scripts (`phase23` through `phase28`) are wired into the driver and the continuous-integration workflow, so the register's diagnostics are regenerated on every push rather than recorded once. The workflow deletes the stored results first, exempting only the two cached Materials Project pulls, which require a credential.

The document compiles with no errors, no overfull or underfull boxes and no undefined references, at 46 pages.

5 Full disposition

Item	Register heading	Disposition
01	Population accounting consistency	Applied, second option
02	Unmatched entries not bounded by 25	Applied
03	Release-based vs confirmed transitions	Applied
04	Gap and cell counts under exclusion	Applied, verified
05	Eligible sample for the IPW rate	Applied, verified
06	Input manifest	Applied, with checksums
07	Sandwich notation	Applied
08	Joint bootstrap, 51 to 53	Applied, verified
09	Cluster pool per analysis	Applied
10	Replicate or seed stability	Applied, seed stability
11	Unreduced compositions	Applied; check not run
12	Extremum selection is descriptive	Applied
13	Exact boolean group definitions	Applied
14	Pseudo-R^2 is not variance explained	Applied
15	Marginal effects: definition and uncertainty	Applied, intervals computed
16	Atomic-fraction interaction model	Applied
17	Additivity constrains, does not require	Applied, heading changed
18	Hydride caption precision claim	Applied, measured
19	U is not a sufficient explanation	Applied
20	Causal verbs	Applied
21	U exclusivity claim	Applied
22	Open-shell wording and copper	Applied
23	Reference convention vs gauge shift	Applied
24	Mean and dispersion both differ	Applied
25	Frozen-4f as a conditional concern	Applied
26	Accuracy does not propagate into disagreement	Applied
27	Shared selection remains unresolved	Applied
28	Winner's curse is a mechanism, not a fact	Applied

Item	Register heading	Disposition
29	No mediation or independence claims	Applied
30	Exact-zero gap provenance	Applied, verified at source
31	IPW is a sensitivity check	Applied
32	Weight and positivity diagnostics	Applied, measured
33	Crystal-system adjustment claim	Applied
34	Pseudopotential exclusion claim	Applied
35	Per-feature attribution of weighting	Applied
36	Entry coverage is a proxy	Applied
37	Class proportions and standardisation	Applied, measured
38	All matched cells in a supplement	Applied, new appendix
39	One-sided bound needs its condition	Applied
40	Known-material margins within 20 meV	Applied, deleted with evidence
41	Unidentified alternative reference set	Applied
42	Threshold definition and pnictide values	Applied
43	Cluster pool for the budget splits	Applied, corrected
44	Upper-estimate language	Applied, and cross-fitted
45	Informal heuristics not evaluated	Applied
46	Catch definition and split variability	Applied
47	Clean regeneration and independent review	Applied in part; review open
48	"Strongest in every specification"	Applied
49	Coverage, metastability, Q1/Q3 framing	Applied
50	Intercept label, plot reference, contrast	Applied
51	Claim-level citation verification	Applied in part; 6 of $\sim$31
52	Abstract and front-matter length	Applied in part